

062400618
2026-06-25 ~ 2026-06-27
7 × NVIDIA A800-SXM4-80GB GPU 1
356 333 Traffic+CRIB 1.2
20242
20 epoch early stopping patience = 5 batch size = 32 Traffic 16/4 lr = 1e-3

1.1 □□□□

Model	SAITS	SAITS+PatchTST	Model
A SAITS	144	137	SAITS+PatchTST
B SAITS	64	59	SAITS+PatchTST
C SAITS	48	48	SAITS+PatchTST
D SAITS	24	21	SAITS+PatchTST
E SAITS	64	56	SAITS+PatchTST
F SAITS	12	12	SAITS+PatchTST
Total	356	333	

1.2 背景

3 21 862

Category	Count	Percentage	Avg Rating	Feedback Summary	7:1:2 Ratio	Action Plan
Weather	21	52,696	10 Avg	Excellent weather conditions overall 21 reviews	7:1:2 Avg	Continue monitoring weather updates
Electricity	321	26,304	1 Star Avg	Poor electricity service 321 reviews/complaints	7:1:2 Avg	Investigate power issues immediately
Traffic	862	17,544	1 Star Avg	Heavy traffic congestion 862 reviews	7:1:2 Avg	Consider alternative routes Electricity 2.7 Avg

□ □

- **Weather** CoIFNet-full

- **Electricity**321 MissTSM-full CRIB-full h336 MSE
- **Traffic**862 CRIB-full batch_size=4504 /epoch

333/356 Traffic862+ CRIB-full CRIB patching + MultivariateNormal 862 batch_size=4 ~72GB epoch 504 8.4 3.5 23 40+

1.3

	CRIB/ColFNet/MissTSM		
	10 epoch, patience=3	20 epoch, patience=5	
	0.1, 0.3	0.5, 0.7	
	3	2	

1.4

CRIB →

	(B, L, C) patching	patch (B, P, N, patch_len) patch_len=12
	Linear	TCNBlockdilation=2^i
	mu + exp(0.5*logvar)*eps	MultivariateNormal(loc, diag(softplus(scale)))
		+ 1% 0.1
	Linear(seq_len, pred_len) + Linear(d_model, C)	reshape (B, N, P*d) → Linear → (B, pred_len, N)
		mask-aware RevIN

CRIB Electricity 39% MultivariateNormal

ColFNet →

|--|--|--|

模块	实现	备注
Backbone	FFN MixerBlock	TSBlock+GELU + AttentionBlock+sigmoid 激活函数
Attention		AttentionBlock + Gated Values
Embedding		<code>nn.Embedding(24, dim) + nn.Embedding(7, dim)</code> 词嵌入/位置嵌入
Normalization	RevIN+norm / denorm+mask	RevON+norm / norm-fore / denorm+mask

数据集 Electricity 占比 5%Weather/Traffic 数据集 -1.2% / +0.4%数据分布不均匀

MissTSM模型结构 → 模型实现

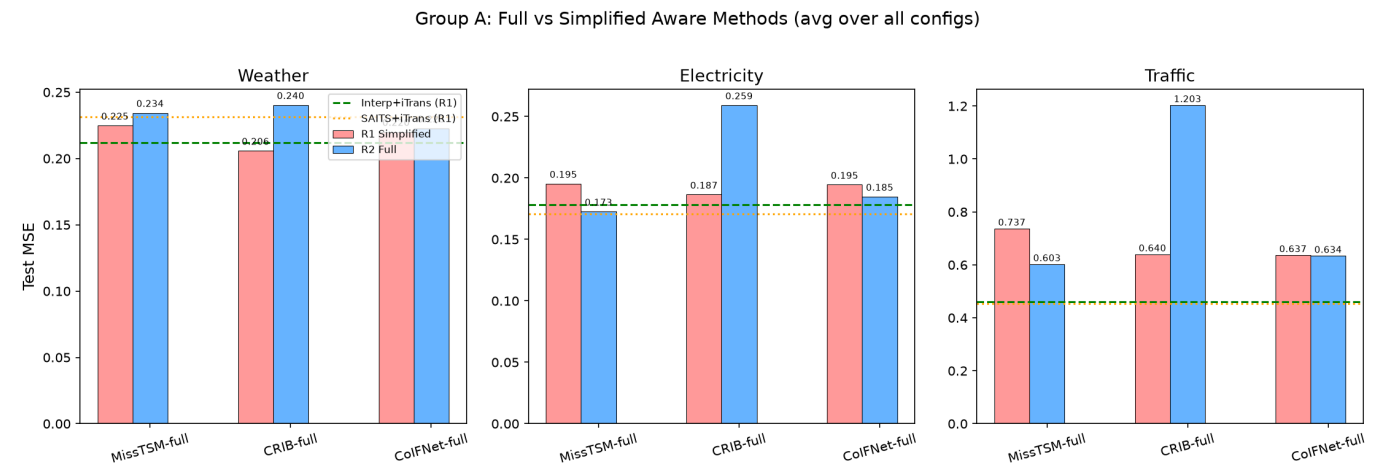
模块	实现	备注
Block	MissTSM	MissTSMSkip+skip connection
Skip connection	<code>x = mask * x_inp + (1-mask) * out</code>	

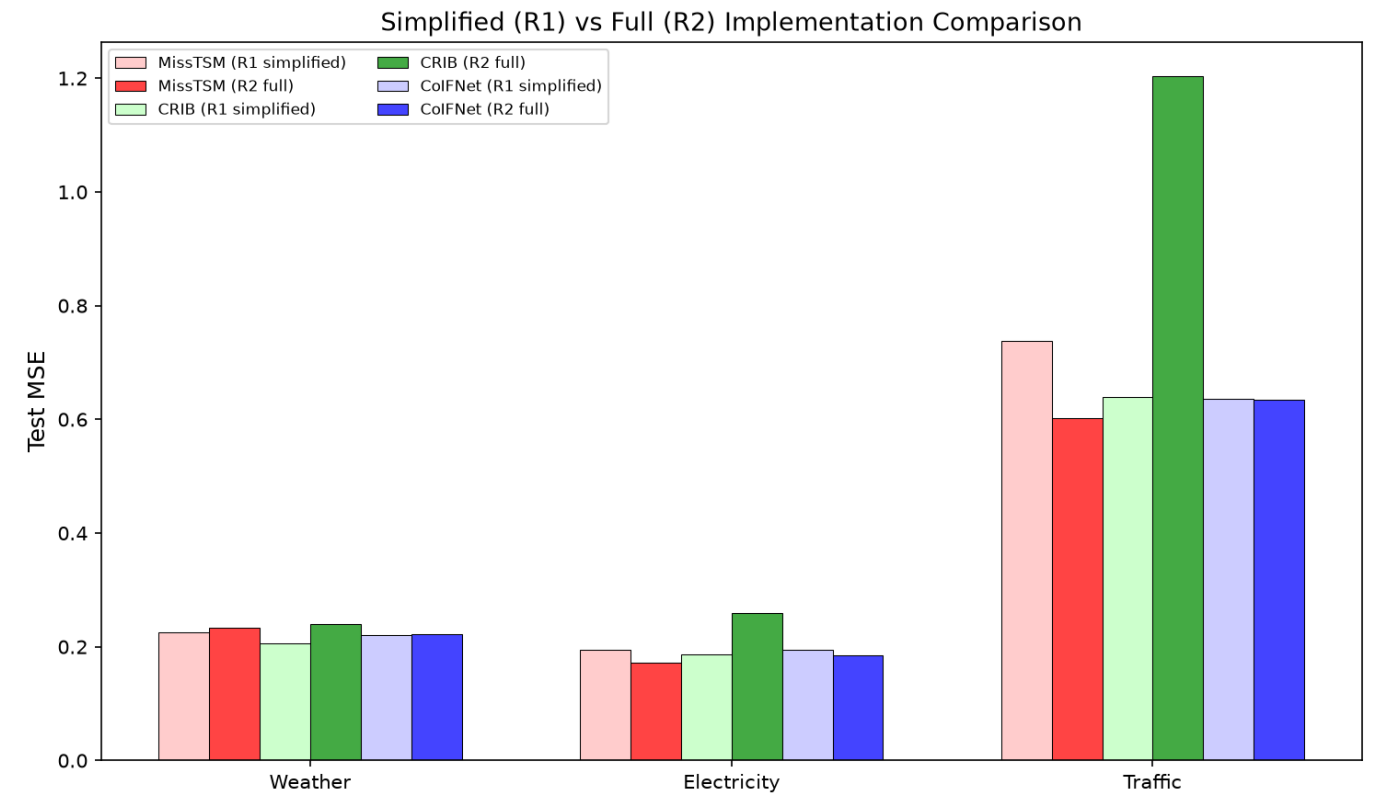
模型 skip connection Electricity 占比 11.6%Traffic 占比 18.3%数据分布不均匀“模型”数据分布不均匀

模型 A——模型 vs 模型

2.1 模型

模型 MissTSM/CRIB/CoIFNet 模型Interp+iTrans + SAITS+iTrans模型





Weather

数据集	比例	数量	Interp+iTrans (R1)	SAITS+iTrans (R1)	MissTSM- full	CRIB-full	ColFNet- full
Weather 1	10%	96	0.1761 / 0.2179	0.1754 / 0.2279	0.1786 / 0.2448	0.1778 / 0.2155	0.1696 / 0.2176
Weather 2	10%	336	0.2746 / 0.2964	0.2918 / 0.3309	0.3000 / 0.3477	0.2952 / 0.3162	0.2804 / 0.3025
Weather 3	30%	96	0.1742 / 0.2187	0.1779 / 0.2425	0.1728 / 0.2389	0.1793 / 0.2192	0.1696 / 0.2177
Weather 4	30%	336	0.2709 / 0.2967	0.2902 / 0.3396	0.2884 / 0.3342	0.3107 / 0.3329	0.2762 / 0.3007
Weather 5	10%	96	0.1755 / 0.2170	0.1676 / 0.2135	0.1783 / 0.2409	0.1691 / 0.2095	0.1667 / 0.2139
Weather 6	10%	336	0.2798 / 0.2988	0.2785 / 0.3062	0.2910 / 0.3351	0.2947 / 0.3158	0.2741 / 0.2990
Weather 7	30%	96	0.1876 / 0.2285	0.1788 / 0.2367	0.1778 / 0.2392	0.1805 / 0.2202	0.1656 / 0.2111
Weather 8	30%	336	0.2887 / 0.3050	0.2898 / 0.3281	0.2877 / 0.3313	0.3146 / 0.3337	0.2788 / 0.3025

Electricity

0000	000	00 00	Interp+iTrans (R1)	SAITS+iTrans (R1)	MissTSM- full	CRIB-full	ColFNet- full
0000 0	10%	96	0.1579 / 0.2521	0.1519 / 0.2500	0.1490 / 0.2515	0.1542 / 0.2496	0.1668 / 0.2709
0000 0	10%	336	0.1873 / 0.2804	0.1807 / 0.2791	0.1877 / 0.2891	0.1839 / 0.2795	0.1941 / 0.2962
0000 0	30%	96	0.1636 / 0.2603	0.1608 / 0.2649	0.1520 / 0.2567	0.1595 / 0.2562	0.1697 / 0.2750
0000 0	30%	336	0.1902 / 0.2862	0.1845 / 0.2894	0.1850 / 0.2892	0.5272 / 0.5265	0.1956 / 0.3001
0000 00	10%	96	0.1648 / 0.2579	0.1567 / 0.2517	0.1547 / 0.2543	0.1545 / 0.2485	0.1648 / 0.2691
0000 00	10%	336	0.1975 / 0.2891	0.1813 / 0.2763	0.1900 / 0.2885	0.1841 / 0.2782	0.1985 / 0.3009
0000 00	30%	96	0.1860 / 0.2760	0.1630 / 0.2625	0.1625 / 0.2653	0.1711 / 0.2626	0.1813 / 0.2828
0000 00	30%	336	0.2200 / 0.3075	0.1878 / 0.2871	0.2014 / 0.3032	0.5397 / 0.5326	0.2086 / 0.3084

Traffic

0000	000	000 0	Interp+iTrans (R1)	SAITS+iTrans (R1)	MissTSM-full	ColFNet-full
00000	10%	96	0.4237 / 0.2954	0.4253 / 0.2940	0.6495 / 0.3792	0.5998 / 0.3285
00000	10%	336	0.4565 / 0.3068	0.4559 / 0.3055	0.6069 / 0.3442	0.6492 / 0.3426
00000	30%	96	0.4373 / 0.3023	0.4374 / 0.2970	0.5775 / 0.3305	0.6068 / 0.3389
00000	30%	336	0.4687 / 0.3135	0.4691 / 0.3105	0.6066 / 0.3372	0.6374 / 0.3477
00000 0	10%	96	0.4455 / 0.3071	0.4296 / 0.2940	0.6162 / 0.3609	0.6101 / 0.3329
00000 0	10%	336	0.4779 / 0.3191	0.4643 / 0.3076	0.6043 / 0.3412	0.6503 / 0.3444
00000 0	30%	96	0.4912 / 0.3294	0.4515 / 0.3001	0.5743 / 0.3287	0.6470 / 0.3519

□□Traffic+CRIB-full □□□□□□□□□□□□batch_size=4□504 □/epoch□□□□□□□□□□□□

Heatmap titled "Full Implementation Improvement over Simplified Version" showing MSE improvement percentages for three datasets (Weather, Electricity, Traffic) across three models (MISSTSM, CRIB, COIFNET). The color scale ranges from -20% (dark red) to 40% (dark green).

Model	Weather	Electricity	Traffic
MISSTSM	-4.3%	+11.6%	+18.3%
CRIB	-16.8%	-39.0%	-88.0%
COIFNET	-1.2%	+5.0%	+0.4%

2.3 A □□□□

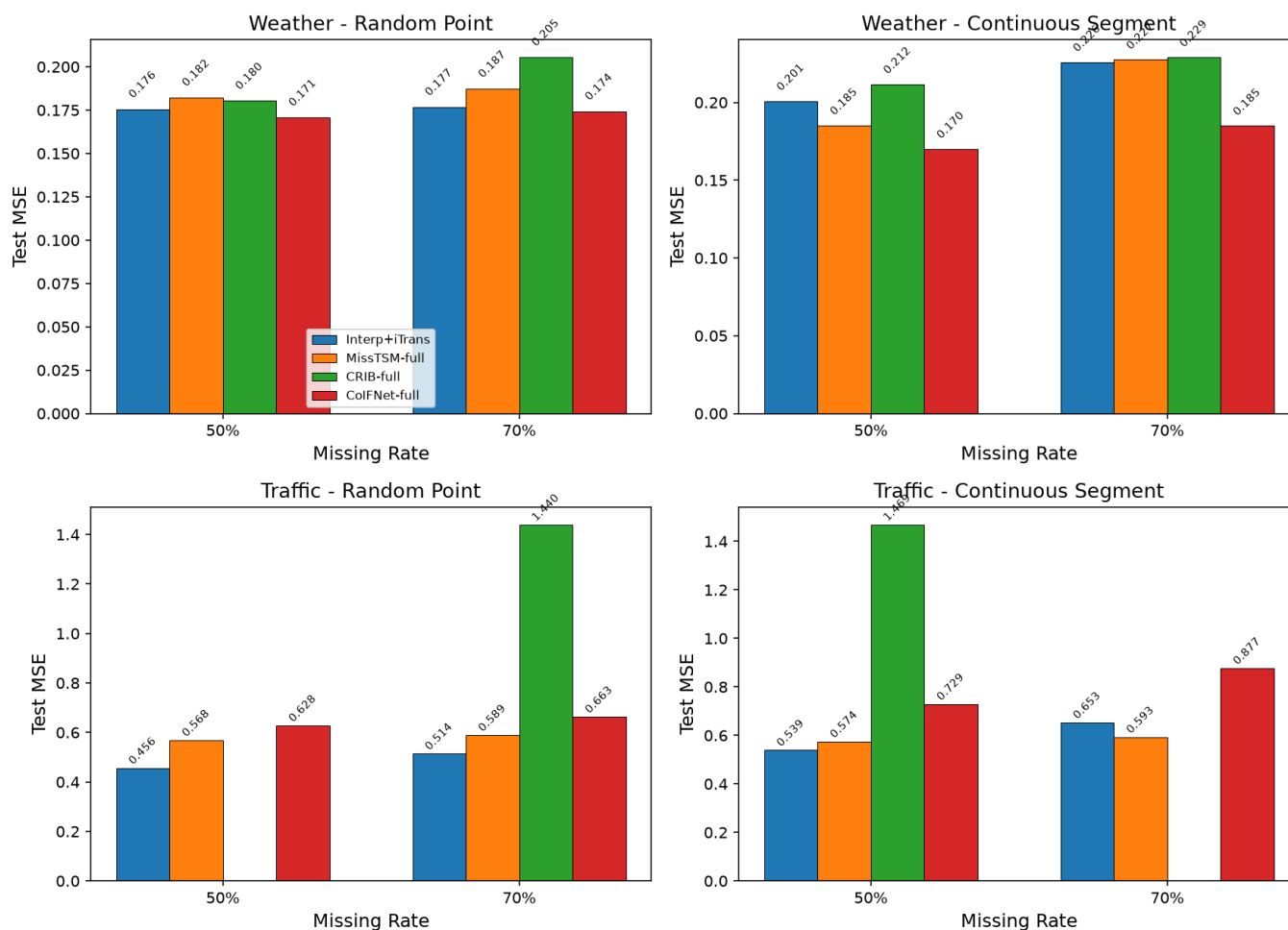
- 6 / 18

2. **CRIB-full** CRIB Electricity -39% MSE 0.53 R1 0.19 MultivariateNormal Traffic+CRIB-full 1.2
3. **CoIFNet-full** Weather -1.2% Electricity +5% Traffic +0.4% CoIFNet
4. Interp+iTrans MSE 0.2834 SAITS+iTrans 0.2851 MissTSM-full 0.3366 CoIFNet-full 0.3473 19-22%

□□□ B□□□□□□□□50%/70%□

3.1 □□

Group B: High Missing Rate Exploration (50% & 70%)

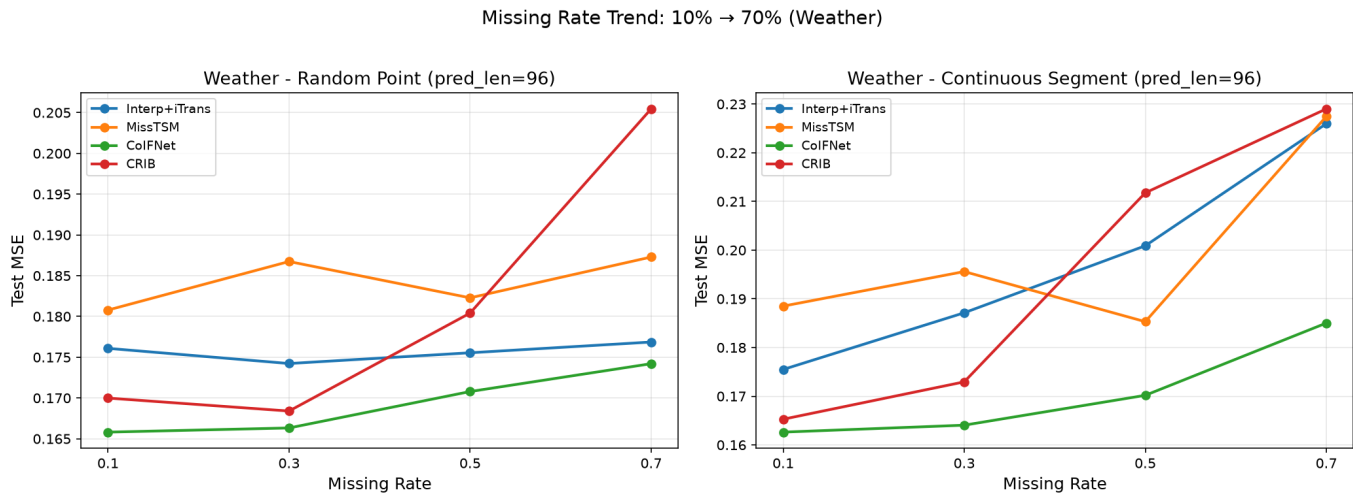


数据集	模型	精度	Interp+iTrans	MissTSM-full	CoIFNet-full
Weather	CoIFNet	50%	0.1755 / 0.2211	0.1823 / 0.2490	0.1708 / 0.2192
Weather	CoIFNet	70%	0.1769 / 0.2258	0.1873 / 0.2521	0.1742 / 0.2215
Weather	CoIFNet	50%	0.2009 / 0.2401	0.1853 / 0.2520	0.1702 / 0.2145
Weather	CoIFNet	70%	0.2260 / 0.2620	0.2275 / 0.2811	0.1850 / 0.2241

数据集	缺失率	比例	Interp+iTrans	MissTSM-full	ColFNet-full
Traffic	0.5	50%	0.4562 / 0.3094	0.5681 / 0.3217	0.6278 / 0.3458
Traffic	0.7	70%	0.5139 / 0.3372	0.5888 / 0.3341	0.6631 / 0.3692
Traffic	0.5	50%	0.5389 / 0.3431	0.5742 / 0.3256	0.7290 / 0.3918
Traffic	0.7	70%	0.6529 / 0.3908	0.5930 / 0.3308	0.8768 / 0.4752

在 Traffic+CRIB-full 数据集上，Weather+CRIB-full 在 B 数据集上 50%: 0.1804, 70%: 0.2055; 在 Traffic 50%: 0.2118, 70%: 0.2290 数据集上 ColFNet-full 数据集上

3.2 缺失率从10% → 70%



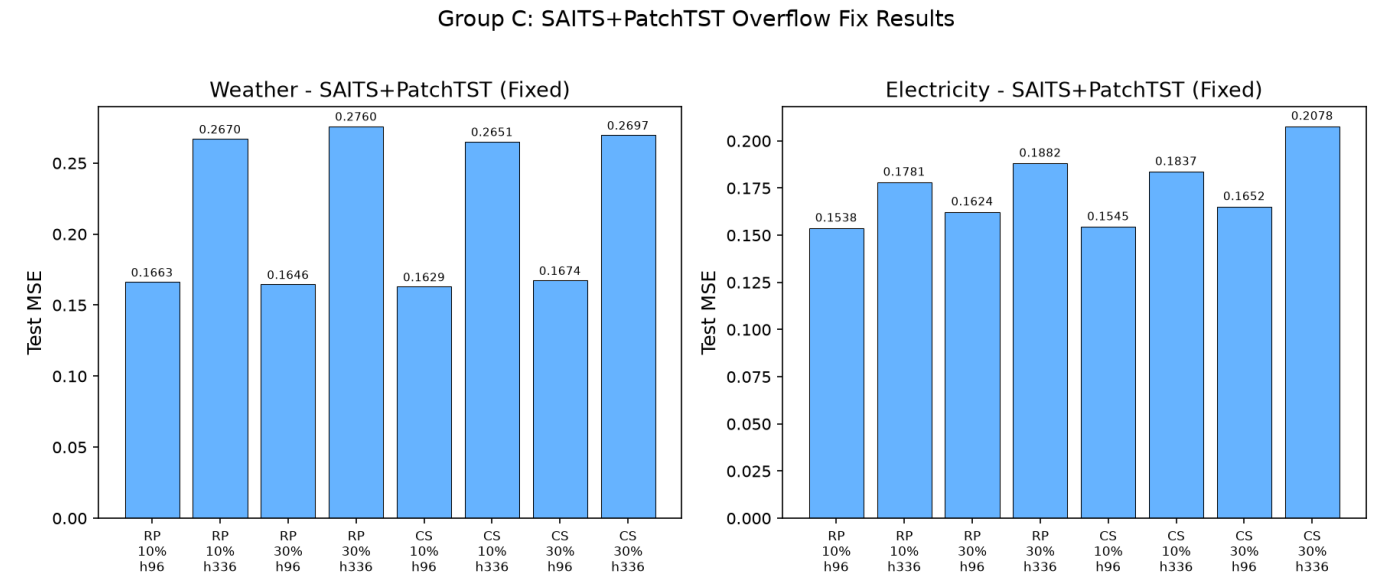
3.3 B 数据集

- ColFNet-full 在 Weather 数据集上** Weather 数据集 4 个数据集上 ColFNet-full 数据集上 MSE 0.1702~0.1850 数据集上 Interp+iTrans 0.1755~0.2260 数据集上 ColFNet-full 数据集上 2.7%~18.1%
- MissTSM-full 在 Traffic 70% 数据集上** Traffic, 数据集, 70% 数据集 MissTSM-full 0.5930 数据集 Interp+iTrans 0.6529 数据集 9.2% 数据集 "数据集" 数据集
- Interp+iTrans 在 Weather 数据集上** Weather 数据集 0.1876 30% 数据集 0.2260 70% 数据集 20.5% 数据集 ColFNet-full 数据集 0.1656 30% 数据集 0.1850 70% 数据集 11.7% 数据集
- CRIB-full 在 Weather 数据集上** CRIB-full 数据集 ColFNet-full 数据集 Interp+iTrans 数据集 Traffic+CRIB-full 数据集
- 在 "数据集" 数据集 Weather 数据集 30~50% 数据集 ColFNet-full 数据集

数据集 C SAITS+PatchTST 数据集

4.1 数据集

SAITS+PatchTST Weather/Electricity 102 20 epoch, patience=548



			H=96 MSE/MAE	H=336 MSE/MAE
Weather		10%	0.1657 / 0.2098	0.2495 / 0.2800
Weather		30%	0.1646 / 0.2136	0.2502 / 0.2809
Weather		10%	0.1643 / 0.2084	0.2480 / 0.2764
Weather		30%	0.1709 / 0.2169	0.2500 / 0.2830
Electricity		10%	0.1481 / 0.2403	0.1734 / 0.2696
Electricity		30%	0.1531 / 0.2487	0.1779 / 0.2737
Electricity		10%	0.1517 / 0.2429	0.1719 / 0.2680
Electricity		30%	0.1640 / 0.2597	0.1930 / 0.2879

4.2 C

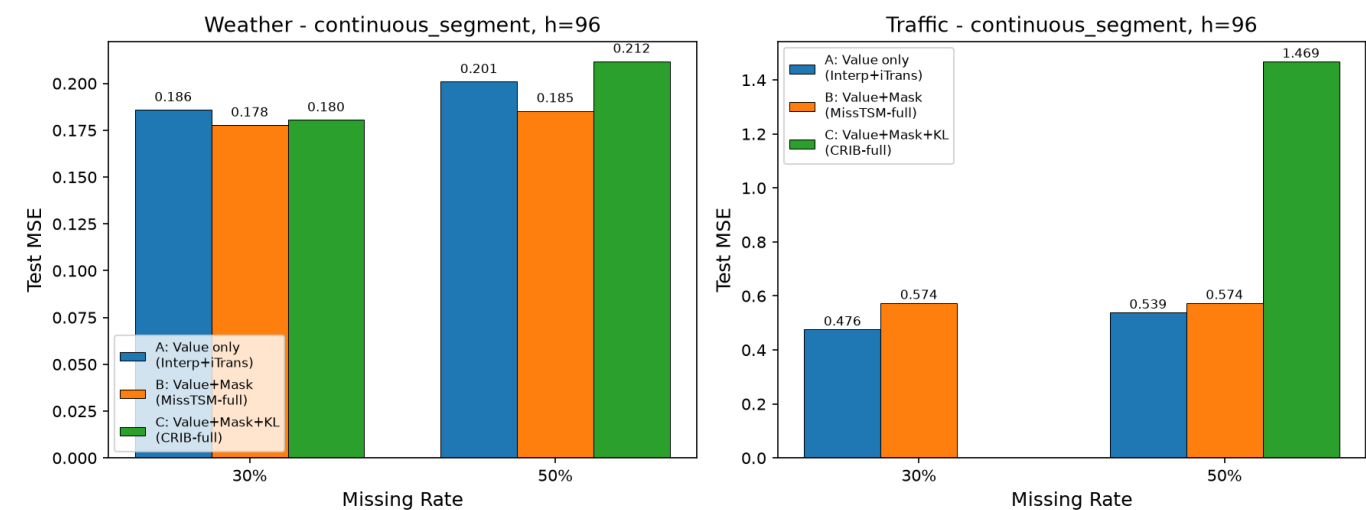
SAITS+PatchTST Electricity h96 MSE 0.1481~0.1640 SAITS+iTransformer MissTSM-full Weather h96 MSE 0.1643~0.1709

D

5.1

""

Group D: Mask Ablation with Full-version Methods



		30% Test MSE	50% Test MSE
Weather	A: Interp+iTrans	0.1876	0.2009
Weather	B: MissTSM-full	0.1778	0.1853
Weather	C: CRIB-full+KL	0.1805	0.2118
Traffic	A: Interp+iTrans	0.4912	0.5389
Traffic	B: MissTSM-full	0.5743	0.5742
Traffic	C: CRIB-full+KL	-	-

5.2 D

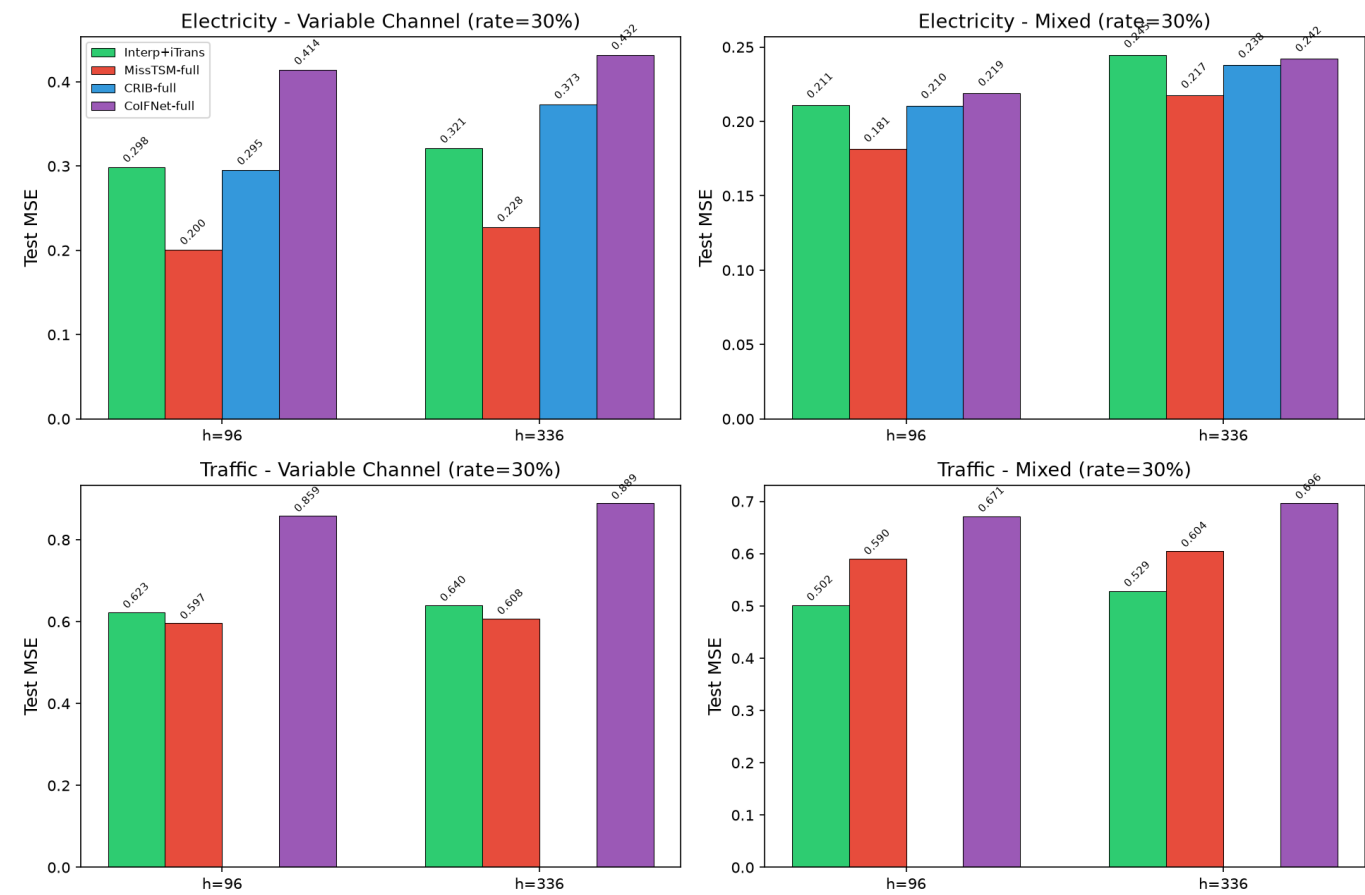
- Weather** MissTSM-full0.1778/0.1853 Interp+iTrans0.1876/0.2009 50% 7.8% MissTSM skip connection
- Traffic** MissTSM-full0.5743 Interp+iTrans0.4912

E

6.1

Electricity Traffic

Group E: Extended Missing Scenarios (variable_channel & mixed)



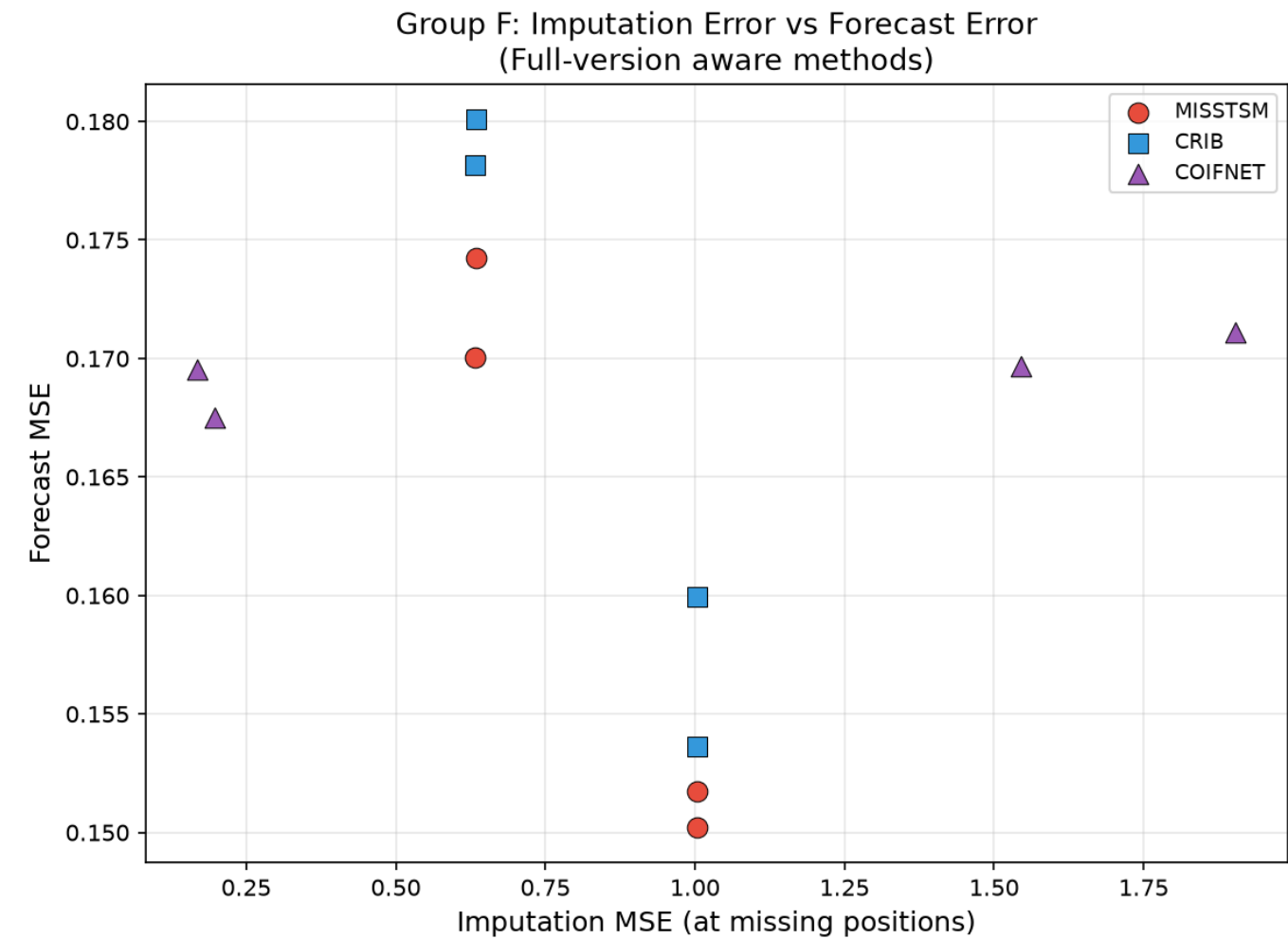
			Interp+iTrans	MissTSM-full	CRIB-full	ColFNet-full
Electricity		96	0.1730 / 0.2673	0.1842 / 0.2777	0.1850 / 0.2776	0.1949 / 0.2940
Electricity		336	0.2060 / 0.2952	0.2154 / 0.3082	0.2182 / 0.3079	0.2290 / 0.3222
Electricity		96	0.1694 / 0.2643	0.1764 / 0.2735	0.1780 / 0.2719	0.1884 / 0.2887
Electricity		336	0.2009 / 0.2922	0.2124 / 0.3044	0.2106 / 0.3046	0.2217 / 0.3179
Traffic		96	0.5030 / 0.3233	0.6078 / 0.3428	-	0.7271 / 0.3800
Traffic		336	0.5260 / 0.3338	0.6119 / 0.3423	-	0.8199 / 0.4221
Traffic		96	0.4860 / 0.3180	0.5984 / 0.3437	-	0.7183 / 0.3863
Traffic		336	0.5191 / 0.3324	0.5999 / 0.3398	-	0.8072 / 0.4239

6.2 E

Interp+iTrans

F

7.1



Model	Dataset	Missing %	Imputation MSE	Forecast MSE
MissTSM-full	Weather	10%	0.6340	0.1742
MissTSM-full	Weather	30%	0.6328	0.1700
MissTSM-full	Electricity	10%	1.0038	0.1502
MissTSM-full	Electricity	30%	1.0038	0.1517
CRIB-full	Weather	10%	0.6340	0.1801
CRIB-full	Weather	30%	0.6328	0.1781
CRIB-full	Electricity	10%	1.0038	0.1536
CRIB-full	Electricity	30%	1.0038	0.1599
CoIFNet-full	Weather	10%	0.1970	0.1675
CoIFNet-full	Weather	30%	0.1687	0.1695
CoIFNet-full	Electricity	10%	1.5454	0.1697
CoIFNet-full	Electricity	30%	1.9034	0.1711

7.2 F

- CoIFNet-full 的 MSE 在 Electricity 上 1.5~1.9 之间 MSE 比 MissTSM-full 的 MSE 大得多 (约 2 倍)
- MissTSM-full 在 Electricity 上的 MSE 约 0.1502 比 CRIB-full 的 MSE 约 1.0 小得多

实验结果

8.1 实验结果

图 A 展示了 Traffic+CRIB-full 和 CRIB-full 在 Weather+Electricity 数据集上的性能对比

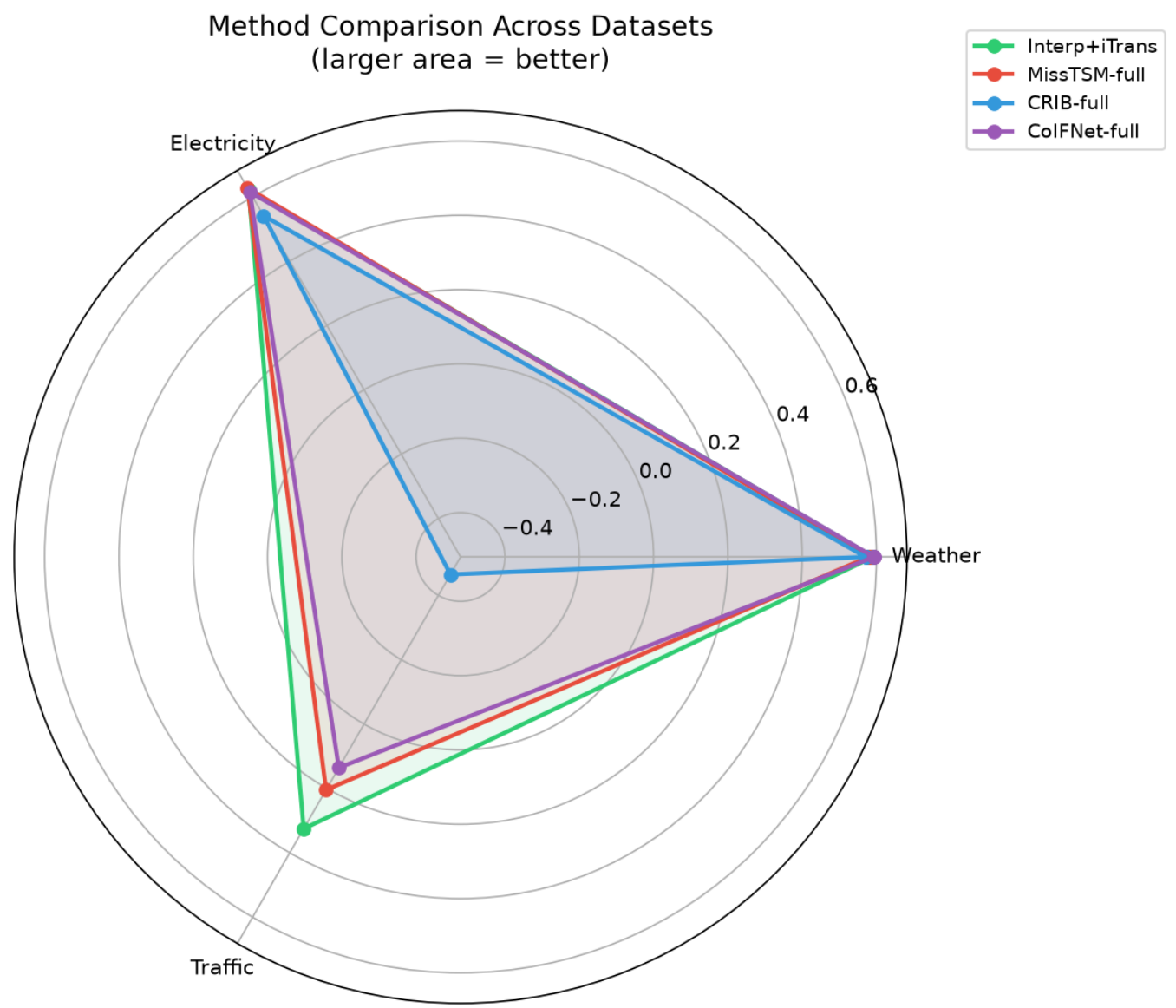


图 A Weather + Electricity + Traffic

Rank	Method	MSE	Relative Change
1	Interp+iTrans (R1 0.2834)	0.2834	—
2	SAITS+iTrans (R1 0.2851)	0.2851	+0.6%
3	MissTSM-full (R2)	0.3366	+18.8%
4	CoIFNet-full (R2)	0.3473	+22.5%

Weather + Electricity CRIB-full

		MSE
1	MissTSM-full	0.2035
2	CoIFNet-full	0.2038
3	CRIB-full	0.2498

CRIB-full Weather+Electricity Traffic 0.4591

8.2

F1: +iTransformer		
F2: >		
F3: MissTSM-full Electricity CoIFNet-full Weather		
F4: ≠	F	
F5: MissTSM Weather Traffic		

8.3

MissTSM/CoIFNet:	MissTSM Electricity 11.6% Traffic 18.3%	
CoIFNet: Weather	CoIFNet-full Weather 50%/70%	
MissTSM: Traffic 70%	Traffic 70% MissTSM Interp+iTrans	
	CRIB-full	

8.4

1. CRIB-full patching + MultivariateNormal + Electricity 39% MSE 0.53R1 0.19 MultivariateNormal Traffic+CRIB-full epoch 504 batch_size 4
2. MissTSM skip connection MissTSM $x = mask * x_{inp} + (1 - mask) * out$ Electricity 11.6%
3. Weather 30~50% CoIFNet-full Traffic CoIFNet-full Traffic

11111

9.1 □□□□

1. **MissTSM** +11~18% **CoIFNet** +0~5% **CRIB** **Electricity** 39% **Traffic**
 - **MissTSM** **CoIFNet** **CRIB** **Electricity** **Traffic**
2. **Interp+iTrans** MSE 0.2834 **MissTSM-full** 0.3366 +18.8%
 - **Interp+iTrans** **MissTSM-full**
3. **MissTSM**
 - **MissTSM-full** **Electricity** 321
 - **CoIFNet-full** **Weather** + 50%/70%
 - **MissTSM-full** **Traffic** + 70%
4. **10 epoch** **20 epoch** **SAITS+PatchTST** 48/48
5. $\text{MAE}_{\text{SAITS+PatchTST}} \times \text{MAE}_{\text{MissTSM}} < 30\% \times \text{MAE}_{\text{MissTSM}} + \text{MAE}_{\text{MissTSM}}$

9.2 □□□□

- 0000000000000000Weather 0000Traffic 00000
- 000000000000000000000000CRIB 00000
- MissTSM 0 skip connection 000"0000"00000000000
- CRIB 000000000000000000000000 negative result 00

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□ □

10.1 Interp+iTrans

[illegible]

```

pipelines.py:TwoStagePipeline → iTransformer → iTransformer
token → itransformer.py:_DataEmbeddingInverted (B, C, L)
nn.Linear(seq_len, d_model) self-attention

```

Transformer self-attention token

[illegible]

10.2 CRIB-full Electricity

□□□□MultivariateNormal □□□□

[illegible]

```
qz = MultivariateNormal(loc=loc, covariance_matrix=torch.diag_embed(scale))
```

loc 与 scale 均为 (B, C, d_model) 其中 C=321 Electricity C=862 Traffic
`torch.diag_embed(scale)` 为 (B, C, d_model, d_model) 表示 Electricity 的 batch 为
 B × 321 × 128 × 128 batch_size=32 为 630MB

在 `crib.py:384` 中 KL 散度

```
kl_raw = kl_divergence(qz_1, prior) # (B, C)
```

`kl_divergence(MultivariateNormal, MultivariateNormal)` 为 $d_{\text{model}} \times d_{\text{model}}$ 的矩阵
 复杂度为 $O(d)$ PyTorch 中 $C=321$ 的矩阵

在 `crib.py:377-380` 中 batch 的 KL 散度

```
x2 = x1 + 0.01 * torch.randn_like(x1)
enc_out_1, qz_1 = self.encoder(x1)
enc_out_2, qz_2 = self.encoder(x2)
```

Transformer 的 CRIBEncoder 在 `crib.py:215-216` 中 reshape 为 (B, P*C, d_model) 的 self-attention Electricity 中 P*C = 8×321 = 2568 个 token 的 attention 为 2568×2568 的矩阵

Electricity h336 的 MSE 为 0.53

h336 与 h96 的 pred_len 在 CRIB 的 Transformer 中 token 为 P*C 在 `crib.py:241-252`

```
self.pred1 = nn.Linear(patch_num * d_model, d_model)
self.pred2 = nn.Linear(d_model, pred_len)
```

pred1 为 $patch_num \times d_model = 8 \times 128 = 1024$ 的 128 的 pred_len=336
 1024 → 128 → 336 的 KL 散度 h336 的 MSE 为 0.53

10.3 MissTSM 的 skip connection

在 `misstsm.py:163-165` 中

```
feat = self.mtsm(x, mask) # 跳过连接
if self.skip:
    feat = mask * x + (1 - mask) * feat
```

skip MissTSMLayer MissTSMLayer
`misstsm.py:105` C " → " → " " "

skip connection backbone iTransformer

Electricity 321 11.6% 30% 321 225 225 96 skip 321

10.4 ColFNet-full Weather

ColFNet

`coifnet.py:296-301`

```
x_full = torch.cat([x_normed * mask, pad_x], dim=1) # (B, total, C)
m_full = torch.cat([mask, pad_m], dim=1)           # (B, total, C)
inp = torch.cat([x_full, m_full], dim=-1)           # (B, total, 2C)
```

30% 70% mask 0 30% 70%
 " / "

Interp+iTrans 70% Weather 21 70% 67
 96×0.7 iTransformer

Traffic

ColFNet backbone SharedModule channel mixing AttentionBlock `coifnet.py:38-81` attention
`hidden × hidden = 128 × 128` temporal mixing TSBLOCK `coifnet.py:21-35`
`total_len = seq_len + pred_len = 96 + 96 = 192`

embedding `coifnet.py:263` `self.embed = nn.Linear(embed_in_dim, hidden)`
`embed_in_dim = 2C + time_feat_proj` Traffic C=862 `embed_in_dim = 1732`
`Linear(1732, 128)` 1732 128 13.5:1 Weather C=21 (42+8):128 ≈ 0.4:1 mask ColFNet

10.5 Weather Traffic

MissTSM `misstsm.py:99`

```
pad_mask = (mask.reshape(B * L, C) < 0.5) # True = = attention
```

Weather C=21 21 30% 15 50% 11

Traffic C=862 30% ~600 MissTSMLayer query token `misstsm.py:70` attention

```
self.var_query = nn.Parameter(torch.zeros(1, 1, q_dim)) # q_dim=64
```

64 query 600 key-value 862 64 query iTransformer attention token

10.6 ColFNet vs

2.2 ColFNet -1.2%~+5%

- 1. TSBlock + AttentionBlock FFN MixerBlock TSBlock GEGLU coifnet.py:32-33 x * gelu(gate) FFN total_len=192 FFN
- 2. Gated Values coifnet.py:69-70, 77 out = out * gates attention head sigmoid attention
- 3. coifnet.py:305 x_mark 8 iTransformer RevIN

ColFNet mask coifnet.py:301

fig1_main_comparison.png	A vs vs
fig2_high_missing_rate.png	B 50%/70%
fig3_r1_vs_r2.png	R1 vs R2
fig4_missing_rate_trend.png	10% → 70%
fig5_saits_fix.png	C SAITS+PatchTST
fig6_mask_ablation.png	D
fig7_extended.png	E /
fig8_error_prop.png	F vs
fig9_improvement_heatmap.png	→
fig10_radar.png	

figures_r2/